Yash Gupta

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Education

Netaji Subhas University of Technology **2021 - 2025**
B.Tech *New Delhi, India*

Delhi Public School, Rohini **2021**
AISSCE (Class XII) - 93% *New Delhi, India*

Delhi Public School, Rohini **2019**
AISSE (Class X) - 90% *New Delhi, India*

Experience

Critical Insights India **Dec 2025 – Present**
Software Engineer *On Site, New Delhi*

- Designed and implemented a **real-time speech-to-text pipeline** for clinical documentation, enabling doctors to dictate notes and receive live transcription updates during patient rounds.
- Developed a **low-latency inference service** that performs continuous speech recognition with intelligent buffering, silence detection, and incremental text aggregation to maintain semantic accuracy.
- Integrated post-processing with **large language models** to automatically refine raw transcripts into structured, professional clinical summaries while preserving medical terminology and numeric accuracy.
- Implemented fault-tolerant persistence for raw audio and session-level recordings to support **auditability, replay, and debugging in production environments**.

Contravault AI **July 2025 – Dec 2025**
Fullstack Developer *On Site, Bengaluru*

- Architected and developed "Tender-Hunt" end-to-end, a **search engine** for tenders that automates scraping tenders from **40+** government websites, ingesting over **1,500** new tenders daily into **AWS Aurora (PostgreSQL)**, using **AWS step functions, ECS**, reducing manual discovery time for enterprise clients from **weeks to minutes**.
- Engineered high-performance search services using **FastAPI** on AWS ECS backed by **PostgreSQL**; utilized Composite **GIN** indexes with **Trigrams (pg_trgm)** and **Full-Text Search** to enable instant fuzzy matching across 200k+ records with sub-100ms latency.
- Built and deployed the customer-facing frontend for Tender-Hunt using **Next.js** and TypeScript, on company product & website, improving website traffic by **50%**.
- Designed a scalable analytics pipeline using **AWS Lambda, DynamoDB & PostHog** to aggregate and store **10+** daily user metrics (e.g., session time, active users, button clicks) across **30+** enterprise clients.
- Collaborated directly with **co-founders** to translate client requirements and stakeholder feedback into actionable technical specifications for the product roadmap.

General Electric **Jan 2025 – June 2025**
Engineering Intern *On Site, Noida*

- Optimized complex **SQL** queries for analyzing large-scale power generation datasets, reducing average query execution time and accelerating the generation of analytical reports for key clients.
- Performed in-depth technical analysis of industrial control system specifications to support bid strategies for projects valued at over **$10M**, ensuring alignment between proposed deliverables and client technical requirements.

Skills

Languages: JavaScript, Typescript, Python, C++
Web Technologies: NextJS, React, NodeJS, REST API, ExpressJS, Selenium
Databases: MongoDB, MySQL, PostgreSQL, Amazon DynamoDB, Appwrite
Developer Tools: Linux, Cursor, Zed, Copilot, AWS, Docker, Postman, GitHub, Git

Projects

Chat with PDF —— RAG, Vector Embeddings, LLMs **Source Code**

- Developed a comprehensive Retrieval-Augmented Generation **(RAG)** system to enable intelligent querying of multimodal documents containing text, tables, and images from scanned PDFs.
- Engineered a data processing pipeline using **FAISS** for efficient vector indexing and similarity search, and Amazon Titan for generating high-quality multimodal embeddings.
- Integrated and evaluated a diverse range of **Large Language Models** through their respective cloud APIs (Google AI Platform, AWS Bedrock) for contextual answer synthesis.

Environment-Initiative-App —— NextJS, MongoDB, React.js, Express.js, Node.js, Solidity, Docker **Source Code**

- Deployed a platform enabling users to create, support, and track initiatives using **MERN stack** and **Web3** for blockchain integration.
- Integrated Web3 to connect to **Ethereum blockchain**, listening for FundsReceived events and updating supporter contribution to any initiative in real-time.

Hackathons

- Ranked among the **top 5 (among 70+ teams)** in the Hack Bake Off Datathon 2024 held at the **NatWest Gurugram** office, demonstrating expertise in **LLMs, data analytics** and problem-solving
- Secured position in **top 15** in NSUT ALPHATHON (competition that enabled competitors to build predictive models using historical market data) organized by WORLDQUANT, thereby winning cash prize.